Recommendations for Implementing Artificial Intelligence Technologies in Nuclear Power Plant Construction Processes

Rev. 1

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Abstract

The report presents application areas for artificial intelligence technologies in nuclear power plant construction based on two complementary models developed by the author: a formal NPP life-cycle model for artificial intelligence and machine learning applications (2025), and a probabilistic optimization and forecasting model for NPP construction projects (2018).

For each application area, the report describes the task, the underlying problem, the proposed solution, and examples of model-based implementation.

The models can be applied during both the preparatory and main construction phases. Implementation is recommended to begin at the project decision-making stage and continue through commissioning. The NPP life-cycle model may subsequently also be used during operation; however, operational applications are outside the scope of this report.

The application areas partially overlap. The same NPP construction task may be considered from the perspective of schedule, supply, requirements, risk, readiness, contracts, or the transition to commissioning. The sections therefore represent different application perspectives within the same integrated modeling framework.

The identified application areas define the potential scope for applying the models and artificial intelligence technologies to NPP construction following an initial screening of possible applications. They do not constitute a final implementation portfolio and do not replace subsequent prioritization.

Contents

Description of the Models

Model A

The 2025 work, *Formalization of a Nuclear Power Plant Life-Cycle Model for AI and Machine Learning Applications*, proposes a unified formal model representing the entire nuclear power plant life cycle, from project conception and engineering through construction, commissioning, operation, and decommissioning, as an integrated interconnected structure.

The model is designed to represent life-cycle processes while accounting for domain-specific requirements and providing a structure suitable for processing by artificial intelligence algorithms.

The model consists of five interacting components:

- **M — Structure.** A multi-level representation of the project that includes hierarchical classification systems for objects and relationships. It distinguishes structural decomposition from dependencies and maintains consistent classifications across hierarchy levels.
- **S — Semantics.** An ontological and linguistic model of the domain. It organizes domain concepts and relation types, term categories, lexical mappings, attributes, and constraints; supports reasoning, consistency checking, and terminology management; and is linked bidirectionally with the structural component M.
- **D — Corpus.** Typed references from model elements to source documents and datasets, providing traceability and supporting evidence-based validation.
- **Q — Quantitative Layer.** Typed attributes and metrics, together with aggregation rules, used to compute readiness, risk, cost, and other quantitative values for analysis and as features for artificial intelligence and machine learning.
- **U — Meta-Level.** An orchestration layer that coordinates M, S, D, and Q. In its machine learning extension, it governs dataset generation, model training, and evaluation while preserving consistency of the overall model.

The key purpose of the model is to create a unified, verifiable, and machine-readable source of structured life-cycle information.

This common foundation can support different intelligent applications, including schedule forecasting, risk identification, readiness assessment, verification of data completeness and consistency, and management decision support.

The concept is therefore not limited to one particular algorithm. It provides a platform model upon which different analytical and AI solutions can be implemented progressively throughout the NPP life cycle.

Model B

The 2018 probabilistic optimization and forecasting model represents a project not as a single linear deterministic schedule, but as a space of possible project execution trajectories.

The project is represented as a directed network of events and activities with branching, alternative scenarios, and probabilistic transitions. Activity duration is represented not by a single fixed value but by a probability distribution. This makes it possible to move from a single planned completion date to a distribution of possible outcomes, to estimate the probability of completing the project by a specified date, and to optimize project execution while accounting for the selected attitude toward risk and multiple project criteria.

The model addresses several interconnected tasks:

1. Identification of multiple critical and subcritical paths under uncertainty through repeated statistical simulation, including estimation of the probability of occurrence of each path.
2. Formalized decision-making at branching points, where alternative project trajectories are evaluated according to project objectives and the acceptable level of risk.
3. Iterative optimization across multiple paths and multiple criteria simultaneously, including schedule, cost, risk, and other project attributes. After the current critical path is optimized, the optimization process proceeds to the next subcritical path and continues iteratively until a conditionally optimal project plan is reached. The optimization is performed in a multidimensional space of project characteristics, with construction of a frontier of feasible solutions.
4. Application of machine learning algorithms to forecast activity duration, classify risks, and detect anomalies using accumulated project data.

Instead of a static project schedule, the approach therefore provides an adaptive system that can update forecasts and optimize the project plan as new information becomes available.

1. Readiness Assessment for Key Project Decisions (First Concrete)

Description.

The preparatory phase includes numerous parallel activities and measures, and the decision to proceed to the main construction phase is one of the most significant decisions in the project life cycle. The IAEA provides a Construction Readiness Review (CORR) service to assess an NPP project's preparedness for construction.

Problem.

Insufficient readiness for the main construction phase can result in schedule delays and cost overruns and create significant risks for subsequent stages of the NPP life cycle.

Solution.

The preparatory and main construction phases are represented within a unified model with formal criteria governing transitions between stages. Based on the interdependencies among preparatory-phase activities and measures, the model calculates the probability of achieving the required level of readiness by a specified date and quantitatively assesses the impact of the current extent of incomplete work on the construction schedule and cost, including the timing of commissioning.

Comment.

The models can be used to substantiate readiness for a key project milestone, support decision-making based on formal criteria, and calculate the consequences of each decision alternative.

Examples of Model-Based Implementation:

- Management of transitions between project life-cycle stages during the preparatory and main construction phases using formal transition criteria.
- A formal Construction Readiness Review framework based on quantitative and qualitative criteria incorporated into the model.
- Aggregated assessment of readiness for First Concrete across all parallel workstreams and activities of the preparatory phase.
- Formal transition criteria, including "proceed," "proceed with documented assumptions" and associated risk assessments, and "do not proceed," supported by modeling of the main construction phase.
- Identification of the minimum sufficient readiness package not only for First Concrete, but also for the first 24 months of the main construction phase, including documentation, contracts, equipment, and work fronts.
- Assessment of the project execution consequences of each decision alternative, including blocked activities, idle time, and affected contracts.
- Dynamic calculation and forecasting of readiness for the key milestone of First Concrete.
- End-to-end scenario analysis. For example, if the main construction phase begins with the current extent of incomplete work, the model can estimate the magnitude and probability of project completion delay and cost overrun.
- Forecasting and optimization of the start date of main construction activities using the current preparatory-phase readiness assessment and a probabilistic threshold.

2. Supply Chain and Equipment Procurement Management

Description.

Thousands of equipment items are supplied for a single nuclear power unit in several sequential procurement and delivery waves. The procurement and manufacturing cycle for long-lead equipment can extend to five years.

Problem.

Misalignment between equipment deliveries and the construction and installation schedule can lead to project delays and cost increases.

Solution.

The delivery of each equipment item is linked in the model to the corresponding installation sequence. A probabilistic delivery forecast is calculated for each item, allowing items at risk of late delivery to be identified before the delay occurs.

The model can account for a wide range of variability and evaluate alternative procurement, delivery, and installation scenarios. Different monitoring and control regimes can be applied to long-lead, medium-lead, and short-lead equipment.

Examples of Model-Based Implementation:

- Synchronization of equipment procurement waves with the installation schedule, financing, customs clearance, and available storage capacity.
- Differentiated supply-chain management regimes, including protected schedule buffers for long-lead equipment, flexible delivery windows for medium-lead equipment, and synchronized delivery for short-lead equipment.
- Backward planning of equipment procurement from the date the equipment is required for commissioning to the required purchase-order placement date.
- Automatic identification of equipment items whose delayed delivery may have a cascading impact on the critical path.
- Monitoring of the readiness of each supply wave and forecasting of expected delivery dates to the construction site.
- Automatic classification of equipment by category, with assignment of standard monitoring regimes and priorities.
- End-to-end control of the complete supply chain, including initial requirements, supplier data, procurement, manufacturing, delivery, incoming inspection, and release for installation.
- Identification of early-delivery risks, including assessment of storage costs, additional handling, and losses associated with congestion at the construction site.

3. Change, Configuration, and Requirements Management

Description.

The configuration of an NPP project comprises the established set of design decisions and supporting documentation. The IAEA defines configuration management as maintaining consistency among design requirements, the physical configuration of the plant, and the associated documentation (IAEA-TECDOC-1335).

Any change may have cascading effects across multiple interconnected project elements.

Problem.

A series of changes, each of which may be individually acceptable, can collectively lead to cost overruns and schedule delays and may ultimately affect plant performance and safety.

Solution.

The model maintains the established project configuration and the relationships among its elements. When a change is proposed, the model automatically identifies the affected elements and assesses the consequences before the change is approved.

Where multiple changes accumulate over a period, the model evaluates their combined effect, which may be several times greater than the sum of the individual assessments.

Examples of Model-Based Implementation:

- Automatic identification of cascading impacts on project elements, including documentation, contracts, activities, and licensing requirements.
- Assessment of the cumulative impact of a series of changes over a specified period on the critical path and total project cost.
- Modeling of consequences under three scenarios: without compensating measures, with compensating measures, and with rejection of the proposed change.
- Two change-control procedures: a full assessment for planned changes and an accelerated procedure for urgent field changes, with the resulting technical debt recorded.
- Management of deviations from the reference design, including verification of compatibility among all introduced changes and with the baseline design.
- Warning when work is being performed using an outdated revision of documentation.
- End-to-end traceability from requirements to implementation and revision control of established design decisions.

4. Design Documentation Management for Construction

Description.

Design documentation for construction should be developed with due consideration of the selected construction methods and installation sequence. Its development also depends on complex processes for obtaining design inputs from equipment suppliers.

Problem.

Documentation is often developed primarily according to the workload of the design organization and the actual availability of input data, rather than in alignment with construction-site needs. As a result, the available lead time between documentation issuance and the start of construction and installation activities is often insufficient.

Solution.

The model integrates engineering, equipment supply, and construction into a single interconnected network in which the critical path is identified and dynamically recalculated.

This makes it possible to prioritize the issuance of each documentation package according to its impact on the construction schedule and to verify the readiness of each project element based on the availability of current documentation revisions.

Comment.

The models can be used to support the timely provision of construction activities with a complete and technically adequate set of design documentation, while taking into account the complex dependencies involved in its development.

Examples of Model-Based Implementation:

- Prioritization of documentation issuance based on construction-site needs.
- Automated generation of documentation issuance schedules linked to the installation sequence.
- Monitoring of the lead time between documentation issuance and the start of construction and installation activities, with early warning when the available lead time becomes critically short.
- Readiness verification based on the availability of current documentation revisions for each project element.
- Warning when work is being performed using outdated or incomplete documentation.
- Identification of missing supplier input data as a cause of delay in the issuance of design documentation for construction.
- Forecasting of documentation reissuance and abortive work resulting from premature issuance based on preliminary design decisions.

5. Licensing

Description.

Licensing and permitting for an NPP project can extend over several years and should be managed as an integrated process covering nuclear licenses, non-nuclear permits, and related approvals.

Problem.

Licensing remains one of the least predictable processes of the preparatory phase. A deficiency in a single licensing package may delay the project for months. For export projects, the challenge is compounded by the need to comply with the requirements of multiple jurisdictions.

Solution.

The model establishes an end-to-end chain linking regulatory requirements, design decisions, supporting justification, and evidence of compliance. At any point in time, it can indicate the extent to which requirements are covered, identify existing gaps, and provide a realistic forecast of licensing and permitting timelines while accounting for the probabilities of alternative process branches.

Reference Example.

In July 2025, Idaho National Laboratory (INL) and Microsoft announced a joint initiative to apply artificial intelligence to accelerate nuclear power plant licensing in the United States. The Azure AI-based system is intended to automate the preparation of engineering reports and safety analysis reports required by the U.S. Nuclear Regulatory Commission (NRC) for construction and operating approvals. This illustrates the practical relevance of AI applications in nuclear licensing.

Examples of Model-Based Implementation:

- Calculation of a regulatory requirements coverage metric based on available evidence of compliance, with automatic identification of gaps.
- Forecasting of licensing and permitting timelines while accounting for probabilistic process branches, including regulatory comments, follow-up inspections, and requests for additional justification.
- Comparison of regulatory requirements from different jurisdictions, together with relevant IAEA standards and guidance, in order to identify conflicts at an early stage.
- An adaptive model configuration for export projects, extended to incorporate national regulatory requirements.
- An integrated licensing and permitting model linking licenses, regulatory reviews, local permits, and infrastructure-related approvals.
- End-to-end traceability of regulatory requirements from the applicable regulatory document to the corresponding design decision, supporting justification, and documentary evidence of compliance.

6. Construction Site and Construction Support Facilities Readiness and Industrialization Methods

Description.

The capacity of the construction support facilities and associated site infrastructure determines the achievable construction rate and should be available by the start of the main construction phase.

An important consideration is the choice between performing installation work directly on site and transferring part of the work to controlled workshop or off-site fabrication conditions, as well as the appropriate degree of prefabrication and modularization. International experience indicates that the greater the degree of prefabrication, the earlier the required fabrication capacity and design documentation for construction must be made available, and the design must accommodate these decisions.

Problem.

The capacity of the construction support facilities may either constrain the construction rate or be excessive, and the required capacity depends on the selected construction methods. Industrialization methods may be chosen without a quantitative assessment of their impact on the project schedule and cost.

Solution.

The capacities of the construction support facilities are linked in the model to the probabilistic construction schedule. The model can identify a balance between the cost of excess capacity and the losses associated with insufficient capacity and resulting idle time.

It can also support selection of the industrialization method by quantitatively assessing the impact of each alternative on the critical path and project cost.

Examples of Model-Based Implementation:

- Balancing the capacities of the construction support facilities against forecast demand across construction phases, with automatic recalculation when the project schedule changes.
- Evaluation of the choice between on-site installation and transfer of work to controlled workshop or off-site fabrication conditions based on their impact on the critical path and project cost.
- Assessment and comparative analysis of the impact of selected construction methods and related design decisions on the required capacities of the construction support facilities.
- Synchronization of the construction support facilities development plan with the probabilistic construction schedule.
- Consideration of seasonal constraints when selecting construction methods and technologies.

7. Transition from Construction to Commissioning and Commissioning Management

Description.

The transition from construction to commissioning is a critical project phase during which construction, installation, and commissioning activities are performed in parallel within the same plant areas and systems. Management of commissioning packages, defined as groups of systems and facilities required for particular commissioning stages, handover of rooms and areas, and the staged progression from Stage A-0 through physical startup, including initial criticality, require complex planning and coordination.

Problem.

Complex coordination and sequencing of activities, overlapping construction, installation, and commissioning work fronts, discrepancies between actual readiness and commissioning requirements, and extensive reliance on temporary systems and configurations can lead to project delays and cost increases.

Solution.

The model optimizes the sequence of construction, installation, and commissioning activities so that complete functional configurations of process systems are established as early as practicable and the readiness of rooms and systems corresponds to commissioning requirements.

The model can also support planning for the timely availability of permanent plant systems and configurations while minimizing reliance on temporary ones.

Comment.

The models can be used to assess the readiness of a functional system and a commissioning package for transition to the next commissioning stage.

Examples of Model-Based Implementation:

- Optimization of the room-handover sequence to enable permanent plant systems to be placed in service by the time they are required for commissioning.
- Formalized criteria for assigning systems the status “functionally ready” based on the completeness of as-built documentation.
- Management of commissioning packages and commissioning stages from Stage A-0 through physical startup, including initial criticality, using calculated readiness metrics.
- A digital readiness passport for each room or area prior to handover for installation and commissioning.
- Backward planning from physical startup through the preceding commissioning stages to the construction and installation schedule.
- Synchronization of construction and installation activities with commissioning activities in order to reduce reliance on temporary systems and configurations.

8. Critical Path Management

Description.

NPP construction schedules are organized as a hierarchy of planning levels. During the main construction phase, the critical path often runs through mechanical and electrical installation activities and piping installation and welding, but it is determined by a much broader set of dependencies, including documentation issuance, equipment deliveries, room handover, defect resolution, and crew availability.

In the model, the critical path is represented as a set of possible critical paths ranked according to their probability of occurrence.

Problem.

A deterministic schedule does not adequately account for uncertainty or hidden dependencies between workstreams, which can amplify cascading effects across the project.

Solution.

The model replaces a single deterministic critical path with a set of possible critical paths, each associated with a probability of occurrence. This makes it possible to allocate schedule contingency selectively to the segments where it is most effective.

When a work front is at risk of becoming unavailable, the model can recalculate alternative execution routes and assess the resulting cascading impacts on other activities and the overall project schedule.

Comment.

The models can be used to identify critical and subcritical paths for project execution, estimate the probability of occurrence of each path, allocate schedule contingency, and assess the consequences of delays.

Examples of Model-Based Implementation:

- Identification of multiple critical and subcritical paths together with the probability of occurrence of each path.
- Targeted allocation of schedule contingency to segments with the highest probability of becoming critical.
- Dynamic routing of mechanical installation activities, including recalculation of alternative work sequences when a work front becomes unavailable and reassignment of crews to available rooms and areas.
- Identification of hidden resource dependencies between workstreams, including shared crews, common suppliers, and embedded items.
- Forecasting the duration of mechanical installation activities based on piping characteristics, crew qualifications, and inspection and quality-control results.
- Modeling of cascading effects with quantitative assessment of how a delay in one workstream affects related workstreams and the overall project completion date.
- Consideration of seasonal and technological constraints in critical-path analysis.

9. Risk Management and Scenario Modeling

Description.

Project risks can interact and produce cascading effects through the network of project dependencies. Quantitative risk analysis can be used to assess schedule uncertainty and to support risk-informed decision-making at key project decision points, including decisions related to project implementation, financing, and the transition to the main construction phase.

Problem.

Methods and tools for advanced project risk modeling, including project and financial risks, remain insufficiently developed. In practice, risk management may either be absent or limited to a superficial level.

Solution.

The model provides a basis for developing advanced qualitative and quantitative project risk assessment algorithms at all levels of management, including the use of artificial intelligence technologies.

Comment.

The models can be used to assess the overall project risk profile and evaluate the robustness of project decisions under adverse combinations of events.

Examples of Model-Based Implementation:

- Assessment and analysis of risks based on their quantified impact on the critical path and project cost.
- “What-if” scenario analysis for key risks, including alternative risk response strategies and assessment of the effectiveness of each response.
- Testing schedule robustness against the simultaneous occurrence of multiple adverse events.
- Quantitative justification of schedule and cost contingencies rather than applying fixed percentage allowances.
- Early identification of risks before adverse consequences occur.
- Automatic recalculation of the project risk profile when underlying project data are updated.
- Testing the robustness of a project decision when the underlying assumptions change.
- Calculation of schedule and cost contingencies based on the current project risk profile.
- Identification of risks affecting a work package, contract, interface between project participants, or external condition affecting project execution.

10. Cost Control, Financing, and Investment

Description.

The cost of NPP construction is substantial. Cost estimates are refined iteratively, from an early indicative estimate to a substantiated cost baseline by the time of First Concrete.

One of the key issues shaping the project financial model is the allocation of investment funding between the preparatory and main construction phases. Insufficient funding of early-stage activities can subsequently result in costly delays during construction.

Problem.

Project cost is difficult to estimate because it is influenced by numerous factors, including project execution conditions, the long construction duration, financing conditions, the contracting strategy, and other factors.

Solution.

The model integrates financial and project data within a unified structure. Project cost estimates are refined iteratively as actual data accumulate, while scenario modeling makes it possible to quantitatively assess alternative allocations of investment funding between project phases.

Comment.

The models can be used to estimate project cost, progressively refine cost estimates during project execution, and manage cost variances.

Examples of Model-Based Implementation:

- Multi-level aggregation of costs across the project structure, with automatic calculation of earned value management (EVM) metrics.
- Iterative development of cost estimates from Class 5 to Class 3, with progressive refinement as additional project data become available.
- Scenario modeling of investment funding allocation between the preparatory and main construction phases, with quantitative assessment of the consequences of each alternative.

11. Repeat-Unit Construction and Transfer of Experience Between Projects

Description.

Repeat-unit construction and the use of standardized NPP designs can reduce construction duration and cost when knowledge and experience are transferred systematically between units, sites, and projects.

Problem.

Knowledge and experience gained during project execution are not always analyzed and transferred systematically between projects.

Solution.

The model accumulates actual data from completed projects within a unified mathematical structure and in the weights of trained AI models. This enables each subsequent project to draw on the quantitative and qualitative experience accumulated across previous projects, including actual activity durations, costs, recurring risks, and their consequences.

Comment.

The models can be used to support the transfer of experience between units, sites, and projects.

Examples of Model-Based Implementation:

- Comparative analysis of actual project performance metrics within a unified data and model structure.
- Transfer of experience between units, where data from the first unit improve the accuracy of forecasts for the second and subsequent units.
- Optimization of parallel construction of multiple units, taking into account shared resources and the optimal time offset between units.
- Accumulation of project experience in the form of trained models.
- Portfolio management of multiple NPP construction projects over a 10–15-year planning horizon.
- Refinement of model parameters by project type, including domestic and international projects.

12. Localization and Import Substitution

Description.

Adaptation of export projects to national requirements may involve two distinct approaches: planned localization or import substitution defined by contractual requirements, and equipment substitution undertaken in response to external supply constraints.

Problem.

Equipment replacement is not simply a matter of substituting one item for another with similar technical parameters. Without end-to-end verification, the consequences may affect multiple interfaces, including regulatory, design, installation, and operational interfaces.

Solution.

When an equipment item is replaced, the model traces the consequences across the entire dependency chain, from licensing requirements to installation interfaces. It also enables quantitative comparison of alternative replacement scenarios in terms of their impact on schedule, cost, and regulatory acceptability.

Examples of Model-Based Implementation:

- Technical and economic justification of localization levels and monitoring of their achievement in export projects.
- Consideration of export-control requirements and equipment availability constraints as factors within the model.
- Identification of single-source equipment items whose procurement or delivery may affect the critical path as supply-chain vulnerabilities.
- Assessment of the acceptability of equipment substitution with consideration of project requirements, installation conditions, licensing requirements, and operational implications.
- Comparison of replacement alternatives by schedule impact, cost, and the extent of required modifications to design documentation.

13. Contract Strategy and Risk Allocation

Description.

The choice of contracting model and the allocation of risks among project participants have a long-term effect on the manageability of the project.

Problem.

Errors in the selection of the contracting model and in the allocation of risks can lead to disputes and increased project costs.

Solution.

The model represents the dependencies among project participants and can provide quantitative support for the development of contract terms and conditions based on the project risk profile and for the allocation of risks among the parties.

Comment.

The models can be used to analyze the structure of contractual relationships, boundaries of responsibility, and how the consequences of deviations are allocated among project participants.

Examples of Model-Based Implementation:

- Quantitative support for contract terms and conditions based on the project risk profile.
- Quantitative support for appropriate allocation of risks among project participants.
- Quantitative support for delay claims, establishing the relationship between the relevant event, activity delays, critical-path impacts, and resulting cost consequences.
- Assessment of contracting strategy for export projects, taking into account localization requirements, financing terms, and procurement rules.
- Development of a map of contractual obligations among project participants, including required delivery dates and revisions of deliverables.
- Assessment of the feasibility of the contractual structure, considering production, financial, and organizational factors.
- Analysis of the causes and consequences of delays for internal analytics and preparation of materials for claims management.

14. Integrated Planning and Schedule Management During the Main Construction Phase

Description.

NPP construction schedules are organized as a hierarchy of planning levels, including the Level 1 directive/overall schedule, Level 2 integrated schedule, Level 3 schedules, and Level 4 operational schedules. Maintaining consistency across these levels is essential for effective project management.

Problem.

Schedules at different planning levels can become misaligned and lose relevance. Unrealistic target dates may also be repeatedly revised during project execution.

Solution.

The model supports scenario and probabilistic analysis in both directions across the schedule hierarchy: from the directive schedule to operational schedules and back, and from the commercial operation date through the commissioning stages to First Concrete and the preceding project-start decisions.

The model can assess the realism and substantiation of schedules at all levels before approval, evaluate alternative execution scenarios, and optimize the schedule across multiple parameters simultaneously.

Comment.

The models can be used to maintain consistency among schedules at all planning levels, assess their realism and substantiation, and support controlled coordination of changes between schedule levels.

Examples of Model-Based Implementation:

- Scenario modeling at key project decision points.
- Aggregation of progress data from lower schedule levels to key project milestones, with identification of deviations affecting the overall project schedule.
- Analysis of delay consequences through recalculation of the critical path and assessment of alternative recovery measures.
- Testing schedule robustness against typical disruptions, including equipment-delivery delays, unavailable rooms or work fronts, rework, and personnel shortages.
- Backward planning from the commercial operation date through the commissioning stages to substantiate the required timing of preceding activities.
- Progressive refinement of the baseline schedule as actual project data accumulate.
- Controlled propagation and recalculation of changes across directive, integrated, and operational schedules.

15. Finance and Investment

Description.

The project financial model should be established at an early stage and updated throughout project execution. The cost of capital remains one of the most sensitive parameters affecting NPP economics, and construction delays can significantly increase the financial burden on the project.

Problem.

Each month of delay generates not only additional direct costs, but also debt service costs and lost revenue. As delays accumulate, the project may approach or exceed the threshold of economic viability.

Solution.

The model automatically recalculates the financial consequences of schedule changes and can identify the threshold beyond which the project may lose its economic viability.

Comment.

The models can be used to assess the project's financial resilience, its financial and economic profile, and the impact of schedule delays on project economics.

Examples of Model-Based Implementation:

- Automatic calculation of the full cost of delay, including site overheads, debt service costs, and lost revenue.
- Optimization of the investment profile and financing schedule based on a probabilistic assessment of construction progress.
- Quantitative support for contract terms and conditions based on the project risk profile.
- Monitoring of the project's proximity to financial and economic risk thresholds as delays accumulate.
- Forecasting of total project cost based on current schedule performance and trends.
- Calculation of the required contingency as project uncertainty progressively decreases.
- Assessment of the impact of delays on financing requirements and the financial resilience of the project.

16. Planning of Room Handover, Work Fronts, and Area-Based Installation

Description.

Completion of the physical building structures does not by itself mean that rooms are ready for installation activities. Readiness also depends on conditions such as finishing work, temperature control, cleanliness, ventilation, and electrical power supply.

Misalignment between the room-handover schedule and the installation and commissioning schedules can have a direct impact on project duration and cost.

Problem.

Project schedules are often developed at the level of buildings and systems, while the physical execution of work takes place within individual rooms and areas. The absence of even one required readiness condition may block an entire work package.

Solution.

The model brings planning down to the room and area level, generates work packages, and verifies that all required readiness conditions are satisfied. If any of these conditions is not met, the model can automatically redirect resources to other work packages that are ready for execution.

Comment.

The models can be used to assess the readiness of a room, work front, or work package before the start of a specific type of activity.

Examples of Model-Based Implementation:

- Formation of area-based installation work packages with verification of all required readiness conditions, including civil readiness, cleanliness, ventilation, access authorization, documentation, and equipment availability.
- Optimization of the room-handover sequence based on the functional logic and interdependencies of plant systems.
- Planning for complete room readiness during construction, including the availability of leak-tight doors, embedded items, and protective linings.
- Selection of the appropriate degree of prefabrication and transfer of work to controlled workshop or off-site fabrication conditions based on the resulting impact on the critical path.
- Monitoring of room readiness using a digital readiness passport for each room.
- Verification of room and system readiness for handover to installation, commissioning, and testing activities.
- Planning of room handover with consideration of interfacing activities and the conditions required for subsequent work.

17. Lessons Learned, Standardization, and Knowledge Repositories

Description.

International experience in NPP construction is systematically documented, but the transfer of this knowledge into the practices of a specific project often remains fragmented. Recurring causes of deviations may therefore need to be identified repeatedly from project to project.

Problem.

Knowledge about project problems and their solutions often remains dispersed across disconnected documents and the individual expertise of specialists. When project teams change or personnel move to another project, a significant part of this knowledge may be lost.

Solution.

The model can automatically identify recurring causes of deviations and link lessons learned to specific elements of the project structure. When decisions are made on a new project, relevant precedents and their consequences can be retrieved and taken into account.

Comment.

The models can be used to support the accumulation, retrieval, and reuse of knowledge, precedents, lessons learned, and the rationale underlying previous project decisions.

Examples of Model-Based Implementation:

- Systematic identification of recurring causes of deviations and subsequent refinement of model parameters.
- Preservation of project experience in the weights of trained AI models.
- Semantic search for relevant precedents and lessons learned to support decision-making on new projects.
- Training of models on data from completed and ongoing projects.
- Retrospective analysis of completed projects to refine forecasting models.
- Identification of similar situations by deviation type, type of work, contract type, and project phase.

18. Internal Control and Project Audit

Description.

NPP construction generates a substantial volume of project data, including design decisions, contractual obligations, acceptance records, test records, and quality-control results. Internal control and project audit should ensure the verifiability of project decisions and verify compliance with applicable requirements throughout all phases of the project.

Problem.

The volume and diversity of project data make internal control labor-intensive and time-consuming. Relationships between decisions, their underlying justifications, and actual implementation results are often traced manually.

Deviations and inconsistencies may therefore remain undetected until their consequences become apparent. Monitoring contractor performance in terms of schedule, quality, and fulfillment of contractual obligations also requires the comparison of large volumes of information from multiple sources.

Solution.

The model structures project data within a unified interconnected system and provides end-to-end traceability from a decision to its justification and result. It can also identify patterns, hidden relationships, and inconsistencies across project data.

This increases project transparency and enables structured project data to be used for systematic internal control and audit preparation.

Comment.

The models can be used to improve project transparency, ensure traceability of project decisions, and support internal control and project audit activities.

Examples of Model-Based Implementation:

- End-to-end traceability of project decisions from the applicable requirement through the supporting justification to actual implementation and documentary evidence.
- Automatic identification of inconsistencies between design decisions, contractual obligations, and actual project data.
- Reconstruction of the complete decision history for any project element, including who made the decision, when it was made, the basis for the decision, and its consequences.
- Verification of the completeness of documentary evidence for key project milestones and phases.
- Identification of patterns in project deviations to support early detection of systemic problems.
- Monitoring of contractor performance in terms of schedule, scope, and quality through comparison of relevant project data.

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15	Finance and Investment	7
16	Planning of Room Handover, Work Fronts, and Area-Based Installation	7
17	Lessons Learned, Standardization, and Knowledge Repositories	6
18	Internal Control and Project Audit	6
	Total	**118**

Appendix 2. Functions by Management Level

S — Strategic, T — Tactical, O — Operational

No.	Application Area	S	T	O
1	Readiness Assessment for Key Project Decisions (First Concrete)	7	5	–
2	Supply Chain and Equipment Procurement Management	2	7	3
3	Change, Configuration, and Requirements Management	3	5	3
4	Design Documentation Management for Construction	1	5	4
5	Licensing	6	3	–
6	Construction Site and Construction Support Facilities Readiness and Industrialization Methods	4	2	–
7	Transition from Construction to Commissioning and Commissioning Management	1	5	3
8	Critical Path Management	2	5	2
9	Risk Management and Scenario Modeling	6	6	1
10	Cost Control, Financing, and Investment	3	1	–
11	Repeat-Unit Construction and Transfer of Experience Between Projects	6	1	–
12	Localization and Import Substitution	5	4	–
13	Contract Strategy and Risk Allocation	4	3	–
14	Integrated Planning and Schedule Management During the Main Construction Phase	3	5	–
15	Finance and Investment	7	2	–
16	Planning of Room Handover, Work Fronts, and Area-Based Installation	1	6	3
17	Lessons Learned, Standardization, and Knowledge Repositories	4	3	–
18	Internal Control and Project Audit	3	6	1
	Total	**68**	**74**	**20**

Note: The same function may be applicable at more than one management level; therefore, the sum of values in a row may exceed the total number of functions within the corresponding application area.

Appendix 3. Functions by Project Management Knowledge Area and Claims Management

The classification follows the Construction Extension to the PMBOK Guide (2016). Claims management is included separately as a supplemental construction management area.

No.	Management Area	Application Areas	Functions
1	Project Integration Management	1, 3, 11, 14, 18	35
2	Project Scope Management	3, 4, 6	19
3	Project Schedule Management	1, 2, 7, 8, 14, 16	44
4	Project Cost Management	10, 15	10
5	Project Quality Management	3, 4, 17, 18	26
6	Project Resource Management	6, 7, 16	18
7	Project Communications Management	4, 17	13
8	Project Risk Management	1, 9	18
9	Project Procurement Management	2, 12	13
10	Project Stakeholder Management	5, 13	13
11	Project Health, Safety, Security, and Environmental Management	5	6
12	Project Financial Management	10, 13, 15	17
13	Claims Management	13, 15	14
	Total		**246**

Note: The numbers in the “Application Areas” column correspond to the numbering of the model application areas in the main body of the report. A single application area may relate to several project management areas and to claims management; therefore, the total number of functions shown in this table exceeds the overall total of 118 functions.

Appendix 4. Functions by NPP Project Management Area

The structure follows IAEA Nuclear Energy Series No. NG-T-1.6, *Management of Nuclear Power Plant Projects*, Section 4.

Section	Management Area	Application Areas	Functions
4.1	Integration	1, 3, 11, 14, 18	35
4.2	Scope	3, 4, 6	19
4.3	Time	1, 2, 7, 8, 14, 16	44
4.4	Cost	10, 15	10
4.5	Quality	3, 4, 17, 18	26
4.6	Human Resources	6, 7, 16	18
4.7	Communications	4, 17	13
4.8	Stakeholders and Interested Parties	5, 13	13
4.9	Risk	1, 9	18
4.10	Procurement	2, 12	13
4.11	Health, Safety and Environment	5	6
4.12	Lessons Learned and Operating Experience	11, 17	12
4.14	Licensing (Nuclear, Environmental and Other)	5	6
		Total	**233**

Note: Sections 4.13 (Radiation Dose and Radioactive Waste Management), 4.15 (Emergency Planning and Response (EPR)), and 4.16 (Security and Safeguards) are outside the scope of this report. A single application area may relate to several project management areas.

Appendix 5. Functions by Sections of the NPP Preparatory Phase Recommendations

Section	Topic	Application Areas	Functions
A	Governmental Infrastructure and Decision to Implement the Project	1, 9, 11	24
B	Licensing and Site	5	6
C	Financing and Investment	10, 15	10
D	Engineering and Constructability	3, 4, 6	19
E	Personnel and Competencies	11, 17	12
F	Configuration and Change Management	3	7
G	Scheduling, Planning, Construction, and Installation Activities	1, 2, 6, 7, 8, 14, 16	49
H	Risk Management	9	9
I	Contract Strategy	13	7
J	Procurement and Equipment Supply Contracting	2, 12	13
K	Site Preparation and Infrastructure	6	5
N	Construction Readiness Review	1	9
		Total	**170**

Note: A single application area may relate to several sections of the Recommendations. Some sections are not shown separately because their content is addressed across several application areas of this report.

Appendix 6. Recommendations for Implementation on an NPP Project

Implementation of the models and artificial intelligence technologies is recommended to proceed in stages linked to the phases of the NPP project life cycle.

Stage 1. Development of an Initial Reference Model Configuration

Development of the methodology and architecture, performance of the required research and development work, and initial training of the model using representative NPP design information, available historical data, standards, and regulatory requirements.

Development of appropriate organizational, governance, and regulatory arrangements for the use of artificial intelligence in NPP construction management, together with the infrastructure required for AI-enabled applications.

Stage 2. Pre-Investment Phase

Deployment and configuration of the model for a specific NPP construction project with the required initial dataset.

Further training using project-specific information, including the regulatory framework of the host country and project contract terms and conditions.

Further development of the methodology and its adaptation and standardization for the specific project environment.

Stage 3. Preparatory Phase

The model is used in the project-specific configuration.

Applications may include assessment of readiness for First Concrete, management of licensing and permitting activities, equipment procurement and supply-chain management, design documentation management for construction, and justification of construction support facilities capacity and selected construction methods.

As project data accumulate, the model uses actual data for calculations and forecasting and may be further trained on those data.

Stage 4. Main Construction Phase

The model is used for schedule and critical-path management, change and configuration management, planning of room and work-front handover, cost control, risk management, and coordination of parallel construction, installation, and commissioning activities.

The model uses actual data accumulated during preceding phases and during the current construction phase.

Further training is performed using actual data from the current project and, where available, from parallel projects.

Stage 5. Commissioning

The model is used for commissioning management, assessment of system and commissioning-package readiness for transition between commissioning stages, and synchronization of final construction activities with commissioning.

The model uses actual data accumulated during all preceding phases.

Further training may be performed using actual commissioning data.

Stage 6. Operation

By the beginning of operation, the model has already been used throughout construction and contains complete project information.

By the end of commissioning, the model can be further trained on actual data and prepared for operational functions, including, for example, ageing management.

The model may also support transfer of experience between sites.

Appendix 7. Use of the 2025 Model Mechanisms

The model components are M (Structure), S (Semantics), D (Corpus), Q (Quantitative Layer), and U (Meta-Level). All five components are used across the 18 application areas.

No.	Application Area	Key Mechanisms
1	Readiness Assessment for Key Project Decisions (First Concrete)	Stage-based filtering of the project structure, bottom-up aggregation of readiness metrics, and semantic transition criteria.
2	Supply Chain and Equipment Procurement Management	Aggregation of delivery timelines across the dependency structure, semantic classification of equipment, and delivery forecasting.
3	Change, Configuration, and Requirements Management	Configuration control through structural constraints, baseline configuration, and assessment of cascading impacts along dependency chains.
4	Design Documentation Management for Construction	Documentation traceability across the project structure, semantic verification of current revisions, and prioritization of documentation issuance according to critical-path impact.
5	Licensing	Semantic traceability of regulatory requirements, evidence chains linked to the document corpus, and requirements-coverage metrics.
6	Construction Site and Construction Support Facilities Readiness and Industrialization Methods	Aggregation of capacities across the project structure, utilization optimization, and traceability of design decisions related to construction methods.
7	Transition from Construction to Commissioning and Commissioning Management	Stage-based filtering, digital room-readiness passports, and formal functional readiness criteria.
8	Critical Path Management	Aggregation of activity durations across the project structure, identification of hidden dependencies, routing, and traceability of completed activities to supporting records in the document corpus.
9	Risk Management and Scenario Modeling	Aggregation of risks across the dependency structure, recalculation of the risk profile, early risk identification, and traceability of risk-related decisions to supporting documentation.
10	Cost Control, Financing, and Investment	Multi-level aggregation of costs across the project structure, iterative cost estimation, and traceability of financial assumptions and justifications.
11	Repeat-Unit Construction and Transfer of Experience Between Projects	Training on data from completed projects within a unified model structure and transfer of model parameters between project configurations.
12	Localization and Import Substitution	Traceability of dependencies across the project structure, semantic comparison of substitution scenarios, and assessment of substitution acceptability.
13	Contract Strategy and Risk Allocation	Analysis of dependencies among project participants through the project structure, quantitative support for contract terms and conditions, and traceability of contractual decisions.
14	Integrated Planning and Schedule Management During the Main Construction Phase	Aggregation of data across structural levels, scenario modeling, and traceability of planning decisions.
15	Finance and Investment	Aggregation of cost consequences across the project structure, investment optimization, and traceability of financial models and assumptions.
16	Planning of Room Handover, Work Fronts, and Area-Based Installation	Verification of combined readiness across the dependency structure and use of a digital room-readiness passport.
17	Lessons Learned, Standardization, and Knowledge Repositories	Model training using datasets generated from the project model, semantic retrieval of precedents, and updating of model parameters.
18	Internal Control and Project Audit	End-to-end traceability of decisions through links to the document corpus, identification of inconsistencies across the dependency structure, and aggregation of completeness and consistency metrics.

Note: The “2025 Model” refers to Yuriy Dmitrishin, *Formalization of a Nuclear Power Plant Life-Cycle Model for AI and Machine Learning Applications*, 2025. DOI: 10.5281/zenodo.18527396.

Appendix 8. Use of the 2018 Model Mechanisms

The 2018 model combines PERT/GERT, Monte Carlo simulation, formal decision-making criteria, optimization, and machine learning.

No.	Application Area	Key Mechanisms
1	Readiness Assessment for Key Project Decisions (First Concrete)	Probabilistic assessment of transition readiness, GERT-based scenario branching, formal decision criteria under risk and uncertainty, and optimization of overlapping activities between the preparatory and main construction phases.
2	Supply Chain and Equipment Procurement Management	Multi-criteria optimization of procurement and delivery alternatives by schedule, cost, and risk; GERT-based branching of supply-chain scenarios; and ML-based forecasting of procurement and manufacturing durations.
3	Change, Configuration, and Requirements Management	Recalculation of the network model when project parameters change and assessment of the combined impact of changes across multiple project attributes, including schedule, cost, and risk.
4	Design Documentation Management for Construction	Integration of engineering and documentation-issuance activities into a probabilistic network model together with construction and installation activities, and optimization of documentation issuance sequences.
5	Licensing	GERT-based representation of alternative licensing-process branches, Monte Carlo assessment of licensing timelines, and comparison of alternative strategies under risk and uncertainty.
6	Construction Site and Construction Support Facilities Readiness and Industrialization Methods	Integration of construction support facilities and site activities within a probabilistic network model, optimization of capacity-related alternatives and required lead times, and assessment of the schedule and cost consequences of alternative construction methods.
7	Transition from Construction to Commissioning and Commissioning Management	GERT-based representation of alternative commissioning sequences, including branches associated with unsuccessful tests, optimization of commissioning sequences, and probabilistic planning of the timely availability of permanent systems and configurations.
8	Critical Path Management	Identification of multiple probabilistic critical and subcritical paths, iterative optimization of critical and subcritical paths, ML-based forecasting of activity durations, and identification of hidden dependencies.
9	Risk Management and Scenario Modeling	Monte Carlo simulation across the project network, probabilistic analysis of critical and subcritical paths, formal comparison of alternatives under risk and uncertainty, and ML-based risk classification and anomaly detection.
10	Cost Control, Financing, and Investment	Assignment of cost attributes to activities and project trajectories, Monte Carlo simulation of cost outcomes, and multi-criteria optimization using schedule, cost, and risk parameters.
11	Repeat-Unit Construction and Transfer of Experience Between Projects	Adaptation of model parameters using empirical data, ML-based refinement using accumulated project data, and probabilistic optimization based on updated model parameters.
12	Localization and Import Substitution	GERT-based branching of equipment-substitution scenarios, multi-criteria comparison of alternatives, and probabilistic assessment of schedule, cost, and risk consequences.
13	Contract Strategy and Risk Allocation	Formalization of alternative contracting strategies as a multi-criteria decision problem, quantitative comparison of risk-allocation alternatives, and selection of alternatives using formal decision criteria under risk and uncertainty.
14	Integrated Planning and Schedule Management During the Main Construction Phase	Probabilistic network modeling of project activities, adaptive recalculation as actual project data become available, iterative optimization of critical and subcritical paths, and ML-based refinement of forecasts.
15	Finance and Investment	Use of probabilistic schedule and cost outcomes as inputs to financial scenario analysis, comparison of alternative project trajectories, and multi-criteria assessment of financial consequences under uncertainty.
16	Planning of Room Handover, Work Fronts, and Area-Based Installation	Representation of room readiness and work-front availability within the network model, dynamic selection of alternative execution routes when work fronts become unavailable, and optimization of room-handover sequences.
17	Lessons Learned, Standardization, and Knowledge Repositories	Adaptation of model parameters using empirical data, ML-based identification of recurring patterns, and refinement of predictive models as additional observations become available.
18	Internal Control and Project Audit	Recalculation of the network model using actual project data, ML-based identification of anomalies and inconsistencies, and comparison of actual project execution with modeled project trajectories.

Note: Yuriy Dmitrishin, *Mathematical Optimization and Forecasting of Nuclear Power Plant Construction Progress using Probabilistic Models and Machine Learning Methods*, 2018. DOI: 10.5281/zenodo.15373676.

Appendix 9. Integration of the Models with AI Technologies

The following AI technologies and supporting capabilities can be integrated with the models:

- **Large Language Model (LLM).** Six levels are considered: base; trained on synthetic domain-specific data; trained on domain data; trained on project data; trained on specific-project data; and refined using expert feedback.
- **Graph Neural Network (GNN).** Four levels are considered: base; trained on synthetic domain-specific data; trained on project data; and trained on specific-project data.
- **Machine Learning (ML).** Classical machine-learning methods.
- **Convolutional Neural Network (CNN).** Convolutional neural-network methods.
- **Retrieval-Augmented Generation (RAG).** Generation augmented by retrieval from a knowledge base.
- **Tools.** Tools invoked or created by LLM-based systems.
- **Agents.** Autonomous LLM-based agents.

No.	Technology	Description	Model A	Model B
1	Large Language Model, LLM (base)	Process orchestration, interaction with the model structure, calculations, document analysis, and report generation.	10+	10+
1.1	LLM trained on synthetic domain-specific data	Training on specially generated data representing typical NPP construction processes and situations.	10+	10+
1.2	LLM trained on domain-specific data	Adaptation to terminology, the regulatory framework, and NPP construction project-management practices.	10+	10+
1.3	LLM trained on project data	Adaptation to data from completed and ongoing NPP projects.	10+	10+
1.4	LLM trained on data from a specific project	Adaptation to the structure, decision history, and accumulated experience of a specific project.	10+	10+
1.5	LLM refined using expert feedback	Refinement of the model based on expert assessments and corrections during use.	10+	10+
2	GNN (base)	Learning from the graph structure and identification of hidden dependencies.	5+	3+
2.1	GNN trained on synthetic domain-specific data	Training on generated project graphs representing typical structures and dependencies.	5+	3+
2.2	GNN trained on project data	Adaptation to typical structures and dependencies identified in completed NPP projects.	5+	3+
2.3	GNN trained on data from a specific project	Adaptation to the graph of a specific project, including its actual dependencies and deviations.	5+	3+
3	Machine Learning (ML)	Regression, classification, clustering, and anomaly detection.	5+	5+
4	Convolutional Neural Network (CNN)	Pattern recognition in monitoring data, time-series analysis, and image processing.	3+	2+
5	Retrieval-Augmented Generation (RAG)	Search and retrieval of information from the document corpus, regulatory framework, and knowledge base.	5+	3+
6	Tools	Execution of calculations, access to data, and interaction with external systems.	10+	5+
7	Agents	Execution of complex multi-step tasks, monitoring, decision-making, and coordination among system components.	5+	3+

Note: The values in the “Model A” and “Model B” columns indicate the approximate breadth of potential applications of each technology within the corresponding model; they do not represent the number of ready-to-use solutions.

Large language models are expected to make a major contribution to overall model effectiveness, particularly when combined with other technologies. Successive levels of LLM and GNN training and adaptation can improve the accuracy and relevance of results. The selected combination of technologies and tools determines the overall effectiveness, extensibility, and functional scope of the model.

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